TP_3V3
HDR-M-2.54_1x1
3V3 1

TP_5V
HDR-M-2.54_1x1
5V_PI 1

TP_PACK+
HDR-M-2.54_1x1
PACK+ 1

TP_PACK-
HDR-M-2.54_1x1
PACK- 1

TP_USB_VBUS2
HDR-M-2.54_1x1
P28 1

TP_USB_VBUS1
HDR-M-2.54_1x1
P10 1

TP_USB_VBUS
HDR-M-2.54_1x1
USB_5V 1

TP_USB_DETECT
HDR-M-2.54_1x1
MOTOR_R_2 1

TP_ROBOT_EN
HDR-M-2.54_1x1
ROBOT_EN 1

TP_I2C_SDA
HDR-M-2.54_1x1
I2C_SDA 1

TP_I2C_SCL
HDR-M-2.54_1x1
I2C_SCL 1

TP_I1C_SCL
HDR-M-2.54_1x1
P27 1

TP_M_L_1
HDR-M-2.54_1x1
MOTOR_L_1 1

TP_M_L_2
HDR-M-2.54_1x1
MOTOR_L_2 1

TP_M_R_1
HDR-M-2.54_1x1
MOTOR_R_1 1

## Pi Zero 2W Header

U6
2.54-2*20 - HAND SOLDER BACK SIDE

| | | | |
|---|---|---|---|
| 3V3 | 1 | 2 | 5V_PI |
| I2C_SDA | 3 | 4 | |
| I2C_SCL | 5 | 6 | |
| MOTOR_L_IN1 | 7 | 8 | IMU_INT1 |
| | 9 | 10 | P10 |
| MOTOR_L_IN2 | 11 | 12 | P12 |
| ENC_L_A | 13 | 14 | PACK- |
| MOTOR_R_IN1 | 15 | 16 | MOTOR_R_IN2 |
| 3V3 | 17 | 18 | P18 |
| P19 | 19 | 20 | |
| SERVO2_SIG | 21 | 22 | P22 |
| P23 | 23 | 24 | P24 |
| | 25 | 26 | SERVO1_SIG |
| P27 | 27 | 28 | P28 |
| ENC_L_B | 29 | 30 | |
| ENC_R_A | 31 | 32 | P32 |
| P33 | 33 | 34 | |
| P35 | 35 | 36 | ENC_R_B |
| ROBOT_EN | 37 | 38 | |
| PACK- | 39 | 40 | P40 |

USBC1
TYPE-C-31-M-12

| | | | |
|---|---|---|---|
| PACK- | A1B12 | GND | |
| USB_5V | A4B9 | VBUS | |
| | B8 | SBU2 | |
| R3 5.1kΩ | A5 | CC1 | |
| PACK- | B7 | DN2 | |
| | A6 | DP1 | |
| | A7 | DN1 | |
| | B6 | DP2 | |
| | A8 | SBU1 | EH 4 |
| R6 5.1kΩ | B5 | CC2 | EH 3 |
| PACK- | B4A9 | VBUS | EH 2 |
| USB_5V | B1A12 | GND | EH 1 |
| PACK- | | | PACK- |

C2
470uF
PACK+ + PACK-

## LCD and TOF connectors

H2
210S-1*6P L=11.6MMGold-plated black

| | |
|---|---|
| 3V3 | 1 |
| PACK- | 2 |
| I2C_SDA | 3 |
| I2C_SCL | 4 |
| | 5 |
| P32 | 6 |

R18 10kΩ

H1
210S-1*6P

| | |
|---|---|
| 3V3 | 1 |
| PACK- | 2 |
| I2C_SDA | 3 |
| I2C_SCL | 4 |
| | 5 |
| R17 10kΩ P33 | 6 |

LCD
Header-Male-2.54_1x8

| | |
|---|---|
| 3V3 | 1 |
| PACK- | 2 |
| P23 | 3 |
| P19 | 4 |
| P18 | 5 |
| P22 | 6 |
| P24 | 7 |
| 3V3 | 8 |

## Battery Protection

BAT
B2P-VH (LF)(SN)
PACK+ 1
PACK- 2

C11 22uF 5V_PI PACK-
C7 22uF 5V_PI PACK-
C8 22uF 5V_PI PACK-
C5 10uF USB_5V PACK-
R4 2Ω
C6 10uF USB_5V PACK-

U7
IP5306

| | | | |
|---|---|---|---|
| | | EP | PACK- |
| USB_5V | VIN | VOUT | 5V_PI |
| | LED1 | SW | L2 1uH PACK+ |
| PACK+ C3 10uF PACK- | LED2 | BAT | PACK+ |
| | LED3 | KEY | SW1 TS-1088-AR02016 R5 10kΩ PACK- |
| PACK+ R2 2Ω C4 10uF PACK- | | | |

## IMU

U4
LSM6DS3TR-C

| | | | |
|---|---|---|---|
| 3V3 | 1 | 14 | I2C_SDA |
| PACK- | 2 | 13 | I2C_SCL |
| | 3 | 12 | 3V3 |
| IMU_INT1 | 4 | 11 | |
| 3V3 | 5 | 10 | |
| | 6 | 9 | |
| | 7 | 8 | 3V3 |

SDO/SA0, SDx, SCx, INT1, VDDIO, GND, GND, SDA, SCL, CS, NC, NC, INT2, VDD

C27 100nF PACK-
C28 100nF 3V3 PACK-

## Speaker connector

H3
210S-1*2P L=11.6MMGold-plated black
SP_P 1
SP_N 2

## Motor driver

R1 10kΩ ROBOT_EN PACK-

MD1
DRV8833PWP

| | | | |
|---|---|---|---|
| ROBOT_EN | 1 | 17 | GND PACK- |
| MOTOR_L_1 | 2 | 16 | AIN1 MOTOR_L_IN1 |
| PACK- | 3 | 15 | AIN2 MOTOR_L_IN2 |
| MOTOR_L_2 | 4 | 14 | VINT C19 2.2uF PACK- |
| MOTOR_R_2 | 5 | 13 | GND PACK- |
| PACK- | 6 | 12 | VM PACK+ |
| MOTOR_R_1 | 7 | 11 | VCP |
| | 8 | 10 | BIN2 C22 10nF |
| | | 9 | BIN1 PACK+ |

nSleep, AOUT1, AISEN, AOUT2, BOUT2, BISEN, BOUT1, nFault

C20 10uF PACK+ PACK-
C21 100nF PACK+ PACK-
MOTOR_R_IN1
MOTOR_R_IN2

## Servo connectors

SERVO1
Header-Male-2.54_1x3
PACK-
5V_PI
SERVO1_SIG

C25 100nF 5V_PI
C26 470uF PACK-

SERVO2
Header-Male-2.54_1x3
PACK-
5V_PI
SERVO2_SIG

## Speaker AMP

5V_PI R7 100kΩ P40

AMP
MAX98357AETE+T

| | | | |
|---|---|---|---|
| DIN | 1 | 16 | BCLK P12 |
| GAIN_SLOT | 2 | 15 | GND PACK- |
| GND | 3 | 14 | LRCLK P35 |
| SD_MODE# | 4 | 13 | N.C. |
| N.C. | 5 | 12 | N.C. |
| N.C. | 6 | 11 | GND PACK- |
| VDD | 7 | 10 | OUTN SP_N |
| VDD | 8 | 9 | OUTP SP_P |
| | | 17 | EP PACK- |

C23 10uF 5V_PI PACK-
C24 100nF 5V_PI PACK-
5V_PI 5V_PI

## Motor encoder connectors

MENCODER_L
210S-1*6P

| | | |
|---|---|---|
| MOTOR_L_1 | 1 | |
| 3V3 | 2 | |
| ENC_L_A | 3 | |
| ENC_L_B | 4 | |
| PACK- | 5 | |
| MOTOR_L_2 | 6 | |

MENCODER_R
210S-1*6P

| | | |
|---|---|---|
| MOTOR_R_1 | 1 | |
| 3V3 | 2 | |
| ENC_R_A | 3 | |
| ENC_R_B | 4 | |
| PACK- | 5 | |
| MOTOR_R_2 | 6 | |

| | |
|---|---|
| TITLE: | Sheet_1 |
| REV: | 1.0 |
| Company: | |
| Sheet: | 1/1 |
| Date: | 2026 |
| Drawn By: | |

EasyEDA